

Research paper

A novel surfing performance quantification system and multi-sensor instrumented surfboard: A proof-of-concept

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ABSTRACT

Despite the last Olympic Games and a recent interest in cutting-edge training, surfers and equipment manufacturers don't have any tool guaranteeing them a consistent Surfer-Board-Wave (the combo) match, mostly due to the incapacity to monitor the mechanical behavior of the board. The aim of this research is to develop methods describing both the combo's performance and the board's mechanical behavior. To do so, an instrumented surfboard is engineered, measuring its rigid-body motions and deformations, synchronized with propositions of performance indicators to describe the performance of the combo.

A surfboard is made with a built-in electronic system governing strain-gauges, IMU (Inertial Measure Unit) and Bluetooth communication. To quantify performance, the notion of Jerk is introduced and applied to the surfer and the board. The muscle activity of the surfer is recorded as a performance indicator. The yaw-rotation of the board is recorded as an indicator of surfing radicality. It is proposed to illustrate these indicators with real condition surfing.

The instrumented surfboard proved proper functioning during testing, and the mechanical behavior of the board's tail was clearly identified. The performance indicators, Board-Jerk, Surfer-Jerk, Muscle Activity and board's Radicality are measured and combined into discriminators propositions constituting a world first attempt of a quantified description of the combo Surfer-Board-Wave performance.

1. Introduction

The practice and sport of surfing started in the pacific islands in the 19th century [1; Zimmermann, 2025]. Since, the gliding supports, or surfboards, never ceased to evolve. Mostly in the hands of self-made craft-mans called "shapers", the boards changed materials, weight, shape, and therefore mechanical properties and performance through years of empirical approach. With the equipment evolving, the sport itself kept changing thanks to technical breakthroughs. For instance, the invention of the "thruster" (3-fins surfboard) or the introduction of composite materials (allowing lighter still strong boards) lets surfers push-out the boundaries of the possible maneuvers [2; Mc-Cagh, 2013]. This raises the question of choosing the right equipment for the type of wave, but also for the surfer's skill level.

To determine which board a surfer should use in given wave conditions, the shapers rely on years of experience and empiric learnings. Mainly due to complex solid-fluid interaction phenomenon and an ever-moving environment, the determinist approach has never been used to

associate a board with a user. The notion of performance based on the combo Surfer-Board-Wave has never been defined, studied or measured. This paper proposes novel tools for quantifying surfing performance, the performance of the combo (Surfer-Board-Wave). It aims to be a proof-of-concept of this new method. To introduce this more deterministic way of thinking about surfboards, this work focuses on engineering and manufacturing an instrumented-board able to measure the rigid-body movements (board accelerations and rotations) and the deformations (board flexion and torsion) during the surfing experience. This data will be added to performance quantifying data: introducing the notion of Jerk to surfing (both Board-jerk and Surfer-jerk), the muscle activity of the surfer and the radicality of the maneuvers. Several combinations of those indicators will be proposed to describe surfing quality. To illustrate the experimental protocol, an instrumented surfer will use the board in real oceanic waves.

So, methods of this paper show three parts:

1. The constants of the experiment: the surfer and the surfboard

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2. The instrumented surfboard, electronic system and calibration
3. Measuring surfing performance

2. Methods

Isolating the variables begins with identifying the constants and ends with measuring their impact on the phenomenon. The novel point of view brought by this work consists of measuring surfing performance by a few indicators used in relation to the variable's measurement.

2.1. The constants of the experiment

To design this new way of thinking surfboard science, the first thing is to identify the factors which will remain constant.

2.1.1. The surfer

The surfer is 1.70 m high and weighs 67 kg. A skill level scale from Wyve [3; Wyve, 2020] is introduced as a way of describing the surfer, illustrated in Table 1. In this work, the surfer skill level is 6.

The surfer's stance is called "goofy" because his standing foot is the right foot. Thus, he's putting this foot forward on the board. If the surfer's standing foot was the left, his stance would be called "regular". The stance is a characteristic inherent to the surfer.

2.1.2. The surfboard

The shape is considered a constant of the experiment.

The board's shape is chosen designed by Johnny Cabianca (DFK 2.0, Cabianca Surfboards, Basque Country Surf Company, Zarautz, Spain) from the board he made for Gabriel Medina, 3 times WorldSurfLeague (WSL) World Champion. The board is 5'11" long (1803.4 mm), 19" wide (482.6 mm) at the widest point and 2 1/2" thick (63.49 mm) at the thickest for 29 L volume and 3.05 kg. This model is designed for a thruster use (three fins), it is displayed in Fig. 1.

2.1.3. The wave

To illustrate the protocol, the instrumented board is tested in real oceanic waves conditions from 0.8 m to 1.5 m in Sables d'Or, Anglet, France. The average vortex ratio of the waves surfed was around 3 [4; Mead, 2001]. To refine the analysis, phases of surfing are defined.

Phases' definition is possible thanks to a video covering of the session and thresholds from the instrumented board data. The board is equipped with strain gauges and an IMU that are used also as threshold sensors. The method covering the installation, use, and analysis of the sensors is described in Sections 2.2.1 and 2.2.2.

The measured data must be paired with a video support of the experiment to facilitate the analysis. A camera man is set up on the beach and is told to record the entire session, before the acquisition starts until after it ends, without any cuts and to always keep the surfer in the frame. The video is synchronized with board data using short and sudden events. For instance, when the acquisition is started, the surfer is told to press on the strain gauge's emplacements. The pushes are both visible on the video and on the board data.

The surfing phase can be decomposed into 6 different phases as shown in Fig. 2.

Table 1

Wyve's surfing level scale (Source: Wyve).

Surfing level	Concrete reference
1	First white-water wave take-offs.
2	First green wave take-offs.
3	First oriented take-offs.
4	First speed generation by compressing and expanding.
5	First functional maneuvers.
6	First tight top turns and maneuvers.
7	Mastering basic maneuvers, first aeriels.
8	Competing level.

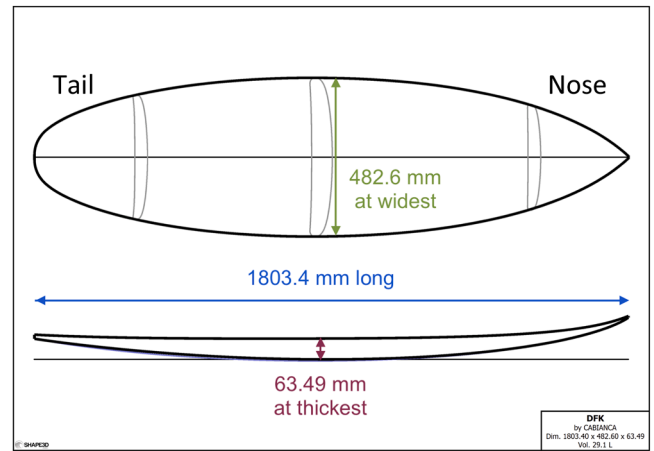


Fig. 1. Plan of the shape used for the instrumented surfboard. The "tail" is the rear part of the board; the "nose" is the front part.

To define the start and end of some phases, the board's rotations, measured by the IMU, are introduced. The axis of the board referential is chosen as presented in Fig. 3. The rotation around the X-axis is the roll, the rotation around the Y-axis is the pitch, and around the Z-axis is the yaw rotation. The IMU gives the rotation speed, noted respectfully R_x , R_y and R_z .

The paddling phase is right before the surfer gets his hands into position for the take-off. The start of the paddling phase is proposed to be 3 s before the take-off.

The take-off is the phase where the surfer gets up on his board. It is defined between the moment the surfer lays his hand on the board, to prepare his push-up; and the moment his hands leave the board, the surfer is up and running.

The trimming phase is a transition phase. The surfer positions himself for the upcoming bottom turn. He uses this time to generate speed, to avoid difficult sections, and to prepare the performance phase of the wave. It starts when the take-off ends, when the surfer's hands leave the board and ends with the beginning of the bottom turn: when the surfer starts rolling the board to orient it toward the top of the wave. This moment t is detected as a sign change of the roll rotation $R_x(t) = 0$.

The bottom turn is a crucial phase to prepare for the next. The bottom turn is the moment the surfer orients his board towards the top of the wave, to hit the top turn. Its intensity can vary depending on the turn the surfer prepares. As defined earlier, it starts with a sign change of R_x and ends with the start of the turn, when the roll change sign again.

The top turn follows the bottom turn immediately. It can be of multiple forms and names: for example, "snap", "carve", "cutback" on the face of the wave or "aerials" over the face of the wave. It is the main performance event of the wave, the maneuver. It begins when the bottom turn ends. It ends whether when the board is changing roll rotation again, for another bottom turn to start, trimming phase or a wave ending; or if the surfer's feet leave the board for a fall.

The finish comes at the end of the wave, when it dies. This phase can also be the fall of the surfer if the top turn isn't fully completed. In competition, the completion of maneuver, even at the dying end of the wave, is rewarded. The finish phase ends when the surfer leaves the board.

It is to be noted that this is a series of phases for a one-turn wave. Longer waves allow the surfer to do multiple maneuvers in a row. In this case, after the top turn phase comes another "trimming/bottom turn/top turn" sequence until the wave dies.

Top level surfers on world class waves can also skip the trimming phase. Right after take-off or a top turn, if the wave offers a top turn section, the surfer can choose to do his bottom turn without repositioning and preparation of the trimming phase. It demands a high level of surfing, a good sense of timing and wave reading, and is not possible



Fig. 2. Phases of surfing on a single maneuver wave.

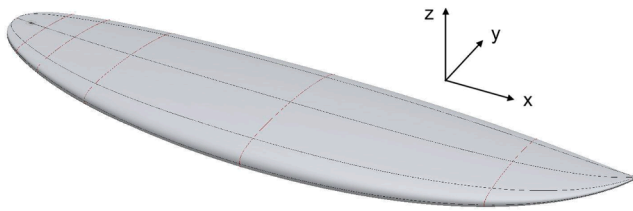


Fig. 3. Board and axis schematics.

on every average wave [5; Piter, 2010 – 6; Malvaux, 2024].

Waves surfed were both “rights” and “lefts”. A wave is called “right” because looking at the beach from the wave breaking point, the wave peels toward the right side. For the same reason, a wave can be called a “left”. This wave’s characteristic is inherent to the swell conditions and the break specifications.

When a goofy surfer is surfing a left wave (like wave 4 and 5), the surfer turns his back to the shore, his toes facing the face of the wave, the surfer is surfing “frontside”. If he was to surf a right wave (like waves 1, 2, and 3), the surfer would turn his back to the wave’s face, his toes facing the shore. This is called surfing “backside”. A regular surfer would be surfing frontside on a right wave and backside on a left wave.

Those two types of surfing are complementary for a surfer’s skills. The two sides demand different pushes, forces, and weight transfers on the board to navigate the board in a similar way [5,6].

2.2. The variables

This paper is focused on building a surfboard measuring its own movement and deformation. The variables to measure are the bending and torsion of the surfboard during the surfing phase.

2.2.1. The electronic system

Electronics are used for developing an in-board measuring and communicating system.

The system is controlled by an Arduino card (Arduino Nano V3 RP2040 Connect, Arduino, Monza, Italy) used also for its internal IMU 6Dof (accelerations and rotation speeds) and its Bluetooth communication [7; Arduino, 2025]. The data is stored in and read from a MicroSD external module with a 16GB SD card. A purple LED is used to warn about the status of the Bluetooth connection. The system is powered by two 3.7 V Li-Po Batteries, rechargeable by its own home-made wireless charger. It has an autonomy of two days with fully charged batteries. A watertight (IP67) push button is the only manual actuator accessible from the outside. It cuts the system from the batteries and redirects them into the wireless charger for the charging phase. It also has a built-in LED used to warn about the on-off state, the acquisition, reading and clearing phase by different blinking speeds.

Finally, 5 strain gauges (SG) are used as the main sensors. A strain gauge is a variable resistor whose value depends on its own deformation [8; Delta lab, 2000].

$$V = V_0 + dV \quad (1)$$

where dV depends on the deformation of its support.

The tension difference read out of the gauge can be calibrated and used to measure in various ways the deformation of the support it is glued on.

To read only the tension difference dV , a Wheatstone bridge is used for each gauge. As dV can be too small for the Arduino to read correctly, an amplifier HX711 is used after each gauge and bridge [8].

Strain gauges are already fully used in various sport equipment’s performance measurements such as cycling [9; Vanwalleghe, 2015], rowing [10; Guseinov 2024], golf [11; Umek, 2017], sailing [12; Pelli- cioli, 2010] or gymnastics [13; Lecocq, 2025].

Computing is used for on-board electronics and data analytics. The onboard script is developed in Arduino and Python 3.8 scripts are used to plot and analyze the data [14; Python, 2025]. A short code is designed to communicate in Bluetooth between the board and a smartphone. The phone application Bluetooth Terminal (Bluetooth Terminal, Lukas Pistrol, Vienna, Austria) is used to send and receive messages. The following functions have been coded (Table 2):

The Bluetooth connection has a 20 m perimeter efficiency outside water, and the acquisition continues even if the connection with the smartphone is lost. The data is then downloaded on the smartphone and can be available for analysis. The whole system communicates by Bluetooth at a 50 Hz rate and gets measurements at a 15 Hz rate. The 15 Hz acquisition frequency is chosen as the maximum the Arduino can handle while writing the data on the SD card. If the frequency is forced over 15 Hz, commands start overlapping and cause frequent corruption in the raw data file. The instrumentation system can measure and transmit data for 6 h straight.

2.2.2. The instrumented surfboard

After a preliminary test on a skateboard proved the system’s proper functioning [15; Ostré, 2025], the instrumented surfboard is then manufactured. The system is fully equipped inside the board, described in Fig. 4 and illustrated in Fig. 5. Only the push-button can be reached

Table 2

Code for Bluetooth communication between the Arduino and a smartphone.

Command sent from the smartphone	Response action from the Arduino
Data	Starts the data acquisition
Stop	Stops the data acquisition
Read	Reads the data from the SD
Clear	Delete the file on the SD
Tare	Sets all the SG data to zero

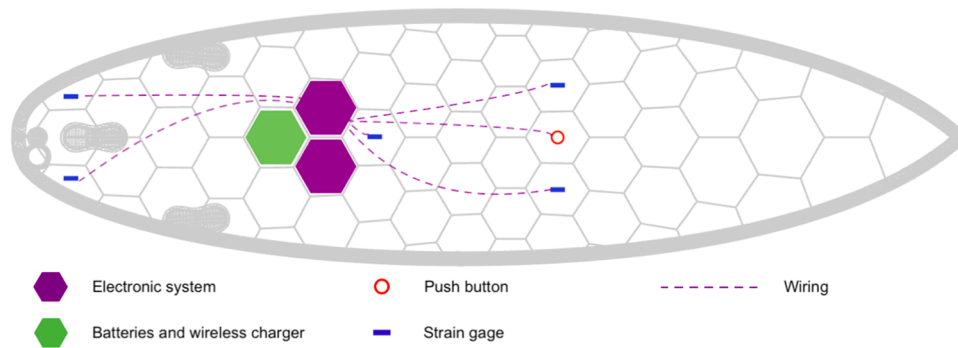


Fig. 4. Strain gages and electronics placement inside the Wyve construction.

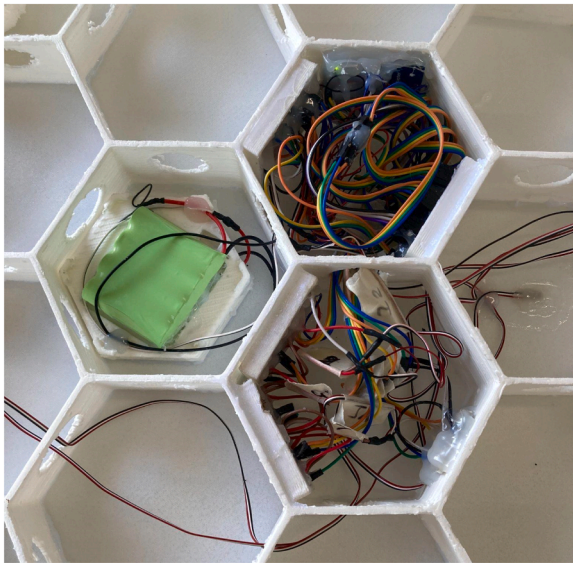


Fig. 5. The electronic system on-boarded into the Wyve construction.

manually from the outside of the device. Both 3D-printing and composite materials are used through the board's manufacturing, with the Wyve technology (4DCore, Wyve, Anglet, France) made of a PLA skeleton core [16; Nunez, 2025] and a fiberglass and epoxy resin lamination. The whole manufacturing has been made by the authors in Wyve's workshop.

Two strain gauges are placed in front of the front foot spot (the middle of the board) and two at the back, behind the back foot spot (the middle of the back quarter of the board). An extra strain gauge is placed between the feet. The push-button pierces the board between the front gauges. The strain gauges are glued with cyanoacrylate and pressed down on the inside of the board for 1 min. Then they are protected with tape and hot glue.

During the experiment, the instrumented surfboard was equipped with standard surfing hardware (wax, fins, pad and leash). A smartphone with a Bluetooth application is used to communicate and store the data coming from the board. Only the instrumented board is tested in the first test.

The test was conducted for 10 min maximum in the water or less if the surfer catches at least two waves and repeated 5 times [17; Lestrade, 2016].

2.2.3. Calibration

Strain gauges give a mV information. To convert this electrical signal to a displacement in mm, the strain gauges system needs calibration. In literature [18; Nunez, 2024], the inversion method is used to get

calibration factors. The strain gauges are solicited separately in torsion and flexion, depending on the case of use. As the instrumented surfboard is the first of the literature, the following calibration protocols are propositions for this specific system, construction and use in real surfing conditions.

Three different static calibration protocols are set up and tested. The first one is a classic 3-point bending calibration, aiming to get calibration factors for each strain gauge and validate the proper functioning in lab conditions. The following two are measuring only the gauges on the tail of the board, considering the bending displacement and torsion of the tail (back-foot position) in comparison to the middle of the board (front-foot position). These two protocols aim to simulate the solicitations of the board in more real conditions, initiated by the feet of the surfer represented by the two clamps.

2.2.3.1. First calibration protocol: three-point bending. The aim of this calibration is to use it as a scale and get the following data thanks to the inversion method: the position in x (length-wise) and y (wide-wise) of the force applied to the board (considered to be the position of the gravitational center of the athlete) and its value [18].

The board is laid down on a three-point bending bench, where the value and position (x, y) of the input force induced by the mass m can be modified. The bench is illustrated in Figs. 6 and 7.

The board's data is recovered using the same method as shown earlier.

The top of the board is stressed at multiple spots (orange points in Fig. 8). For each point, two force values have been applied. The localization of each spot is voluntarily chosen on 3D-printed parts of the board (and not on the hexagon center), so the very localized mass does not break the composite material and spread more reliably through the board.

To get three values (x, y, and F) from the measurements with the inversion method, only three gauge data inputs are needed.

The board data is treated with a Python script testing all triplets of three gauges out of the five. The calibration factors are calculated using the inversion method and finally the precision of each triplet is calculated using validation measurements.

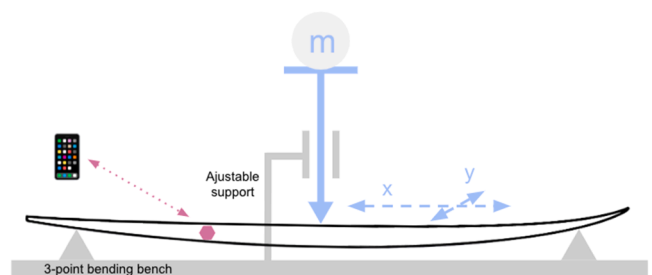


Fig. 6. Schematics of the three-point bending bench and the board.

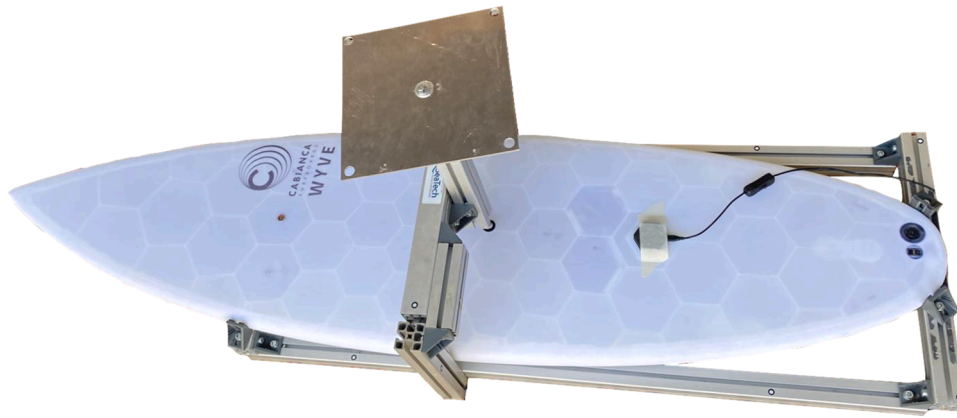


Fig. 7. Picture of the three-point bending bench and the board.

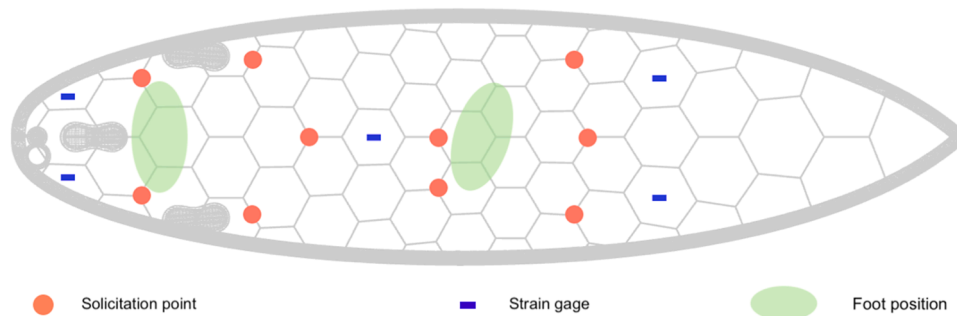


Fig. 8. Solicitation area on the board.

2.2.3.2. Second calibration protocol: tail flexion with IMU reference. A second protocol is proposed with a focus on the tail section. When standing up on the board, the surfer uses the tail of the board to maneuver it, also helped by the fins situated under his back foot. This back part of the board is considered the most important regarding maneuvering, while the front is mainly useful to support the weight of the surfer when lying on the board and paddling [2]. This is why it is proposed to focus on the two rear gauges giving information about tail deformation during surfing.

The following protocols, in flexion and torsion, simulate the real solicitation on the board by the surfer's feet while surfing.

The middle section of the board is fastened to a table and cannot move. A clamp is fastened around the tail of the board where a weight can be hung, as shown in Fig. 9.

On top of the tail is glued an IMU (MetaMotionS, MBientlab, San Jose, USA). The reference angle measured with the IMU is obtained using the accelerometer and the gyroscope (both at 100 Hz frequency) and combined with a Kalman filter (Kalman gain is 0.95) [19; Nasser H, 2014 – 20; Brookner, 1998].

The board data is recorded while the actual bending angle of the tail is recorded with the previous IMU. The board's data is recovered using the same method as shown earlier. The MetaMotion data is recovered using the MetaMotion smartphone app. Both data are plotted and

analyzed with Python.

Data recording on this bench highlighted that the two rear gauges behave differently in torsion and flexion, as illustrated in Fig. 10.

The flexion part of the solicitation can be isolated by applying the following treatment to those two gauges:

$$\frac{C_1 F_1 + C_2 F_2}{2} = \alpha \quad (2)$$

where C_1 and C_2 are the calibration factors, F_1 and F_2 the gauges values.

2.2.3.3. Third calibration protocol: tail torsion with IMU reference. As seen previously, flexion can be isolated using (2). By subtracting the strain gauge values instead of adding them, we get the flexion behavior of the tail. It gives us the following:

$$\frac{C_1 F_1 - C_2 F_2}{2} = \beta \quad (3)$$

The same bench is used for tail torsion than the previous one in tail flexion. A de-axed weight can be hung on the clamp around the tail of the board where a de-axed weight can be hung. In this case, the de-axed length is 500 mm and the weight is 10 kg. A pillar is placed under the clamp, so the board doesn't bend when stressed while its triangular shape allows tilting, shown in Fig. 11.

On top of the tail is glued an IMU (MetaMotionS, MBientlab, San Jose, USA). The reference angle measured with the IMU is obtained using the accelerometer and the gyroscope (both at 100 Hz frequency) and combined with a Kalman filter (Kalman gain is 0.95).

The board's data is recovered using the same method shown earlier.

2.3. The result: measuring surfing performance

The board's data needs to be linked to performance factors that quantify the quality of surfing or the efficiency of the board. As

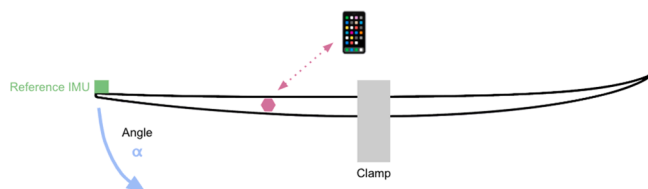


Fig. 9. Schematics of the tail bending bench and the board.

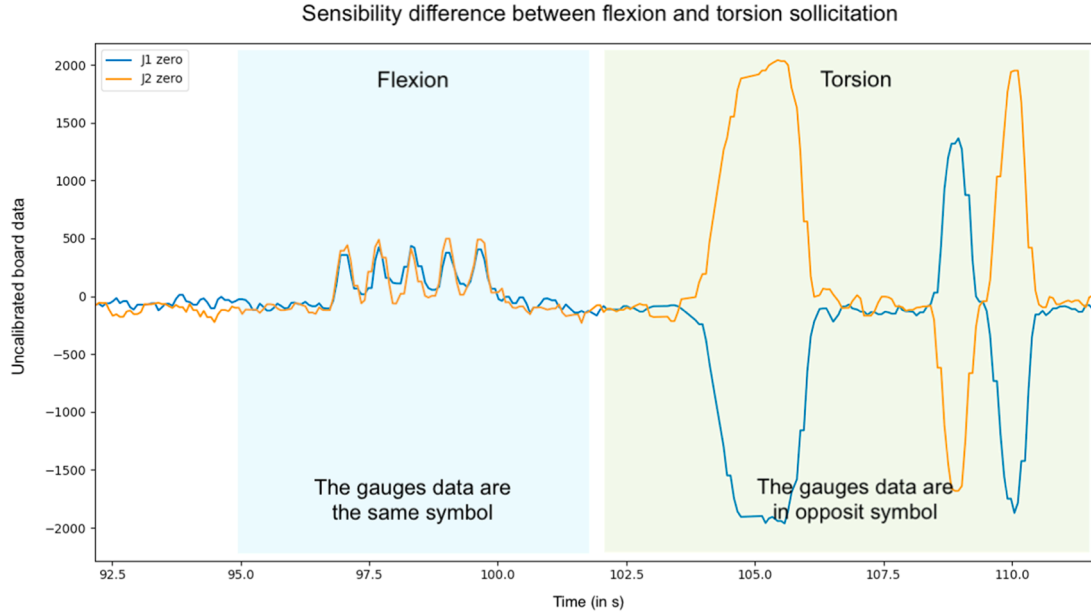


Fig. 10. Sensibility difference of the instrumented surfboard between flexion and torsion sollicitation.

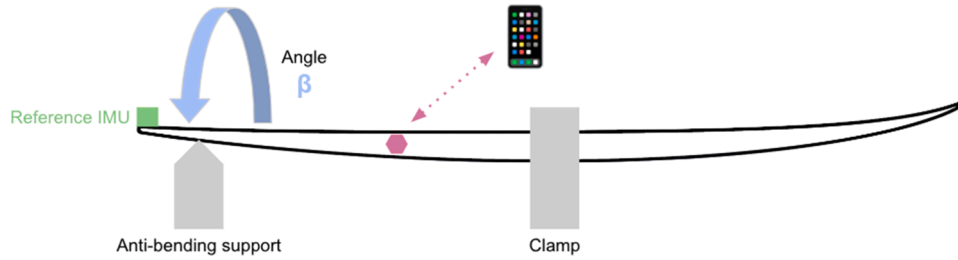


Fig. 11. Schematics of the tail torsion bench and the board.

competitive surfing scores are subjective judging averages, no quantitative performance factor is for now used. It is proposed here to introduce four indicators.

2.3.1. The jerk

The notion of jerk is used to measure the smoothness of a movement. It is the rate of change of acceleration. The smoothest movement is the one with minimal jerk.

The jerk is already being used in several other domains and sport studies. It is frequently used in robotics to find the most efficient trajectory of an upper limb, mimicking the natural refinement of a movement by learning and experience [21; Hogan, 1984]. Following the same logic, it is used in sports to find the smoothest trajectories, non-dimensionalizing the jerk cost with the length and duration of the move [22; Murphey, 2009 - 23; Seifert, 2014].

We use a criterion J that is the normalized integral of the mean-square of the jerk.

It is calculated as followed:

Let $a(t) \in \mathbb{R}^3$ be the acceleration vector at a moment t of an object.

We define the jerk j being:

$$j = \frac{da}{dt} \quad (4)$$

The criterion to be minimized J is defined as so:

$$J = \int_0^T C \frac{j^2}{2} dt \quad (5)$$

where j denotes the jerk, used as its mean-square $\frac{j^2}{2}$ and T the duration of the wave.

To compare the different waves and surfing, the integral is normalized by C that makes J dimensionless. C is chosen as so:

$$C = \frac{T^5}{D^2} \quad (6)$$

where D is the length of the overall trajectory, measured from the integration of acceleration:

$$D = \sqrt{\left(\int_0^T \int_0^T a_x dt \right)^2 + \left(\int_0^T \int_0^T a_y dt \right)^2 + \left(\int_0^T \int_0^T a_z dt \right)^2} \quad (7)$$

The duration of the wave is measured between the time the surfer starts the take-off (pushes up) to the moment his feet leave the board.

2.3.1.1. Board-jerk. The jerk of the board is calculated with the acceleration of the embedded IMU of the Arduino, situated at the gravitational center of the board.

2.3.1.2. Surfer-jerk. The legs and lower body position were measured with inertial units (Cometa WaveTrack Cometa Italy, Milan, Italy). IMUs were calibrated using standing T-pose acquisition [24; Cordillet, 2019]. Fig. 12 shows the IMUs fixed to the surfer.

The acceleration of the IMU situated on the lower back is used to calculate the jerk of the surfer. IMU data is filtered using a 4th order

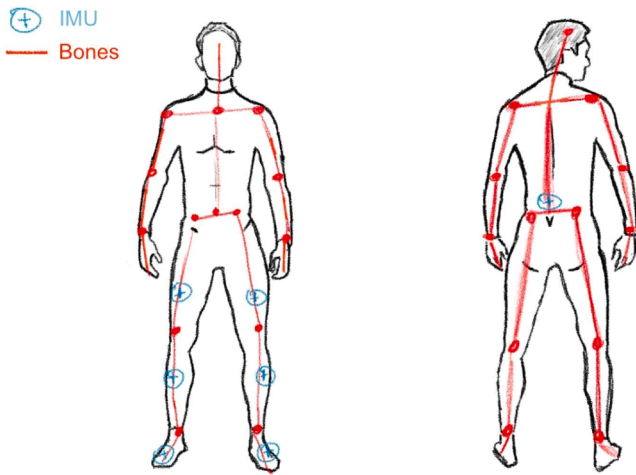


Fig. 12. Schematics of the IMU placement on the surfer.

Butterworth lowpass filter with a cutoff frequency of 7 Hz [25; Csákvári, 2025 - 26; Algoridin, 2025].

2.3.2. Muscle activity

It is considered that relaxed and flowless surfing of a combo Surfer-Board-Wave is proof of a good match of the combo, and so an indicator of performance as one of the competition criteria [27; WSL, 2025]. It is proposed to measure the muscle activity of the surfer, letting the optimum trajectory on the wave be the one minimizing unwanted moves and thus optimizing muscle activity. In literature [25; Csákvári, 2025], minimizing muscle activity is used in a compromise with another performance indicator such as speed, acceleration, or rotation speed.

The surfer is instrumented with EMGs (Cometa MiniWave Infinity, Cometa Italy, Milan, Italy) at a sample rate of 2000 Hz. The 2000 Hz frequency is chosen as it has been used in other sport science research with same hardware [24,25]. SENIAM references are followed to place the electrodes after skin preparation. The following muscles activity was recorded, being the known muscles mainly used in the surfing phase [28; Ruess, 2013]: Trapezius Descendens (upper), Triceps brachii, Biceps femoris, Gastrocnemius Medialis, illustrated in Fig. 13.

EMG data is filtered using a 4th order Butterworth filter with a cutoff frequency between 20 and 400 Hz. A Root Mean Square (RMS) algorithm (Matlab 2023, MathWorks, Natick, MA) with a sliding window of 300 ms was used to smooth the data [26].

To quantify the amount of muscles activity during an entire wave or only a surfing phase, the treated EMG data is integrated on the corresponding time span:

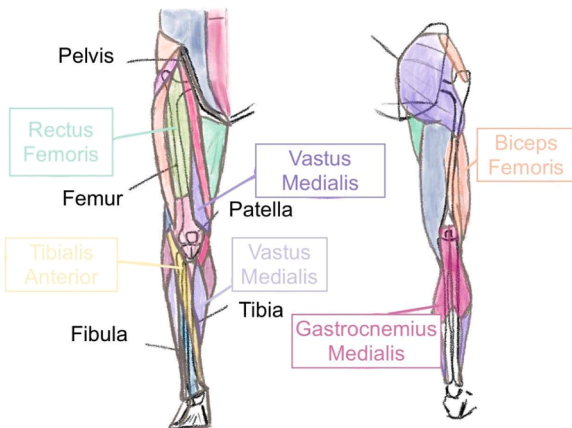


Fig. 13. Schematics of the leg muscles and bones.

Let A_{muscle} be the treated EMG data of a couple left-right of a muscle. It is the sum of the two side muscles. There is one for each instrumented couple of muscles.

$$A_{muscle} = A_{right\ muscle} + A_{left\ muscle} \quad (8)$$

The performance indicator Σ used is calculated as so:

$$\Sigma = \int_0^T (A_{TD} + A_{TB} + A_{BF} + A_{GM}) dt \quad (9)$$

where T is the duration of the considered phase.

2.3.3. Maneuvers and notion of radicality

Jerk and muscle activity are quantities to minimize as indicators of surfing performance. But a quiet activity of a surfer on a wave, keeping his muscle activity and sudden moves to the minimum, would not be considered as optimal performance because only a straight linear trajectory would be described. To get the best performance in sports, the jerk and muscle activity must always be paired with another performance indicator, such as the duration of the move, for instance [25].

Both rotation speed and sweep angle of a surfing maneuver are proposed as performance factors as the brutal changes of trajectories in the top part of the wave (during the turn) are rewarded in competition [27].

To change direction quickly, those radical turns are made following the z-axis of the board, the yaw rotation. The rotation speed R_z and rotation angle θ_z of the maneuver is measured from the embedded Arduino's gyroscope's z-rotation, shown in Fig. 3.

The angle θ_z is measured like so:

$$\theta_z = \int_{t_0}^{t_1} R_z dt \quad (10)$$

Where t_0 and t_1 are respectively the time of the start and the end of the top-turn maneuver. They are determined as the moments the board changes the rotation direction along the X-axis. Also written $R_x(t_0) = R_x(t_1) = 0$.

2.4. Data and scripts availability statement

Raw data samples of board data, IMU and EMG are provided by the authors and can be found in addition to this paper.

Arduino, Matlab and Python scripts used for the treatment and analysis of the data are available from the corresponding author upon request.

3. Results

3.1. Measuring board's flex during surfing

3.1.1. In-lab calibration of the board

3.1.1.1. First calibration protocol: three-point bending. On the three-point bending bench, the calibrated data give the correct weight and position with that average precision: $Dx = 1\%$; $Dy = 13\%$; $DF = 13.5\%$

For instance, where the verification test [790 mm, 97 mm, 20 kg] (meaning 20 kg localized on the point (x, y) where $x = 790$ mm and $y = 97$ mm) is applied on the board, the calibration gives the calculated result $[759.5 \pm 7.6$ mm, 171.8 ± 22.3 mm, 21 ± 2.8 kg]. The respective precisions are shown in Fig. 14.

This calibration allows the correct reading of any input within the solicitation zone. It validates the proper functioning of the board in laboratory conditions.

To use the board in real conditions, it needs another calibration without constant boundary conditions both on the nose and tail of the

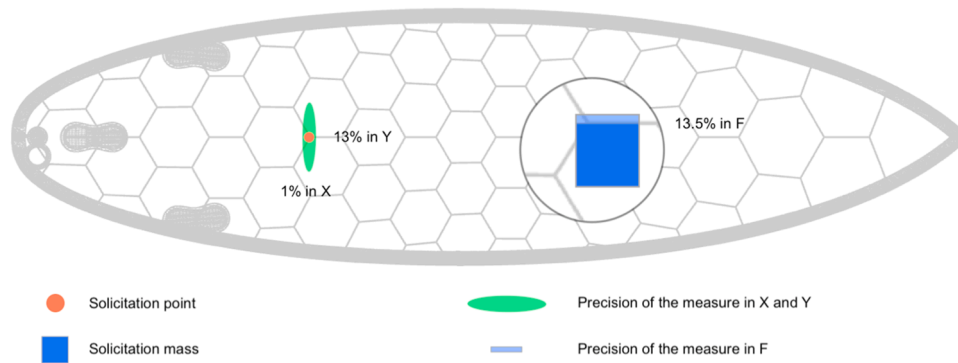


Fig. 14. Result precisions of 3-point bending calibration.

board.

3.1.1.2. Second calibration protocol: tail flexion with IMU reference. The board gives a calibrated angle value that matches the real angle with a 24.6 % precision. For instance, the validation angle measured -0.75° by the IMU, gives $-0.91 \pm 0.18^\circ$ from the board's data. Measurements from the board and the IMU are displayed in Fig. 15.

3.1.1.3. Third calibration protocol: tail torsion with IMU reference. On average, the board gives a calibrated angle value that matches the real angle with an 8.9 % precision. For instance, the validation angle measured -0.97° by the IMU, gives $-1.0 \pm 0.09^\circ$ from the board's data. Measurements from the board and the IMU are displayed in Fig. 16.

3.1.1.4. Calibration recapitulation. The three calibration protocols can be recapitulated in Table 3:

3.1.2. In-situ results

All waves presented here are single maneuver waves, mostly because the in-situ experiment takes place on French beach-breaks where the waves are short.

3.1.2.1. Instrumented surfboard qualitative data. The instrumented surfboard showed quantitative results which allows identification of the different phases in a surfing session, such as take-offs (Fig. 17). Duck-

dives are visible on the gauges data and the paddling cycle visible in the built-in Arduino's IMU (Fig. 18).

3.1.2.2. Instrumented surfboard: measurement of the mechanical characteristics. As shown earlier in Fig. 2, we can note different frames while surfing. The maneuver or top turn is the main performance event of the wave. The bottom turn is a key moment setting up the top turn. Both phases are inseparable as the top turn trajectory depends on the bottom turn trajectory.

For 5 waves, 3 backside and 2 frontside, tilt angle and flexion displacement of the tail is measured during bottom turn and top turn. The performance factors cited in Methods 3. are also measured for waves 4 and 5.

In the following Figs. 20 and 21, the curves displayed are the slices of the board. The blue one is the middle of the board, where the front foot stands. The red one is the tail of the board, where the back foot stands. They are clearly displayed in Fig. 19. The visible tilt angle and the bending displacement of the slices are amplified by 4 in the graph for a better reading; the noted angle and displacement are the original measurement. The slices are seen from the nose perspective.

The mechanical behavior of the board has been measured on 5 waves.

It is possible to visualize the board data along a wave, as shown in Figs. 22–26.

To help with the understanding of the visualization of Figs. 22–26, a description of the waves is given.

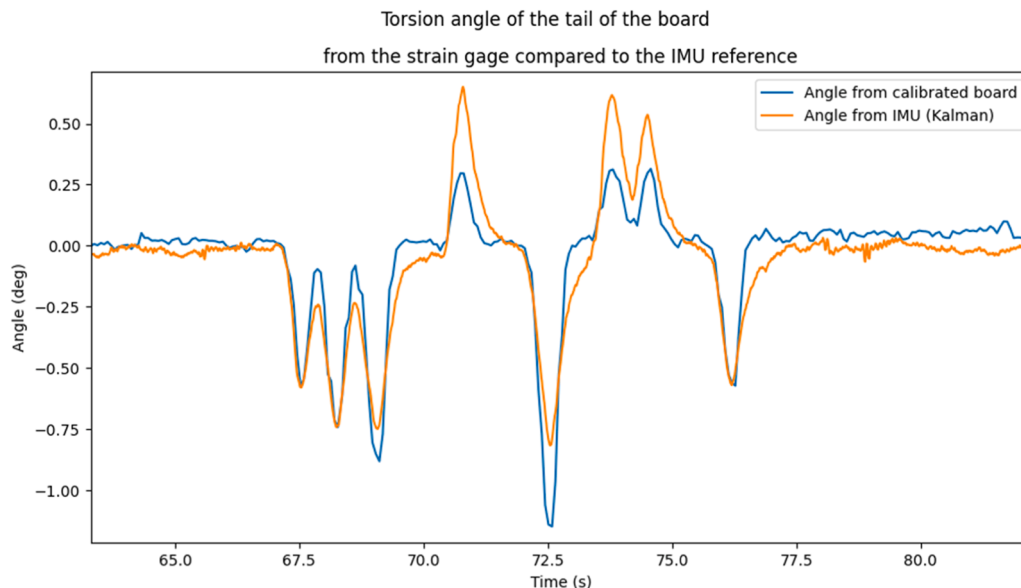


Fig. 15. Calibrated bending angle from the instrumented board and the reference from the IMU.

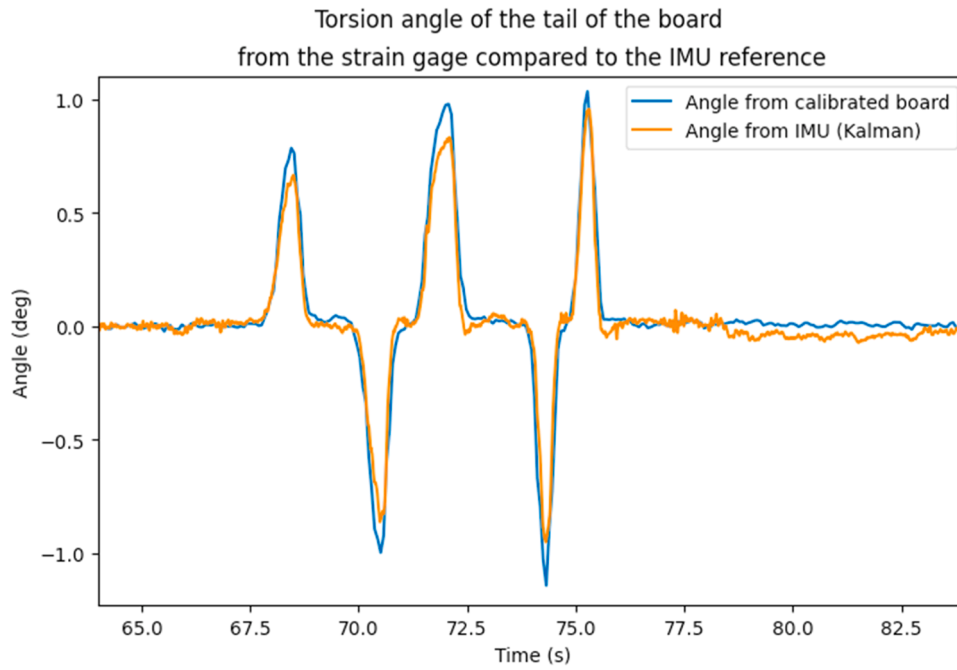


Fig. 16. Calibrated angle from the instrumented board and the reference from the IMU.

Table 3
Precision results for each calibration protocol.

Strain Gages Calibrations Recap	Flexion		Torsion
	1. 3-points bending	2. Clamp bench	3. Clamp bench
Considered strain gauges Measured values	Three, as one measure tool The applied force intensity and its position x, y, F	Only the rear two The flexion bending angle of the tail section in comparison to the middle section α	Only the rear two The torsion tilt angle of the tail section in comparison to the middle section β
values symbols	x: 1 %y: 13 % F: 13.5 %	α : 24.6 %	β : 8.9 %
In-lab Precision	No	Yes	Yes
Usable in real condition	No	Yes	Yes

- Wave 1 is a short right wave (4.5 s). The surfer stands backside on the wave as it is a right. The surfer quickly after the take-off project toward the wave top to land and complete a smooth top turn known as a “floater”.

- Wave 2 is a 4.6 s right wave. The wave peels slowly, enabling the surfer to complete a backside “carve” (long and smooth turn) before diving under the closing wave.

- Wave 3 is 6.6 s right wave. The surfer uses the closing section to complete a backside “re-entry” (quick and vertical turn).

- Wave 4 is a short left wave (5 s). The surfer uses the trimming phase to get himself in position for the turn, a frontside re-entry ending with the fall of the surfer.

- Wave 5 is longer left (7.7 s). The trimming phase is also longer because the surfer must reconnect with the wave rhythm and avoid falling sections. The main turn is a completed frontside carve.

Waves 1, 2, and 3 and waves 4 and 5 were recorded during two uninterrupted sessions. The instrumented surfboard can record multiple waves in the same session without fail in batteries or data storing.

3.2. Performance indicators result

Four performance indicators have been proposed in this work. The

jerk of the board, the jerk of the surfer, the muscle activity of the surfer, and the radicality of the board’s rotation.

In this section, the performance indicators are compared and arranged together in different models proposing performance discrimination tools.

3.2.1. Muscle activity results

Muscle activity measurements alone give results about the efforts of the surfer.

The proportions of the cited-before muscle activity (%), considering the phases of wave n°4 are: Paddling (2.4 s) 41.3 %. Take-off (0.5 s) 15 %. Trimming (2.2 s) 25 %. Bottom turn (0.5 s) 7.5 % and Top turn (0.8 s) 11.2 %. They are displayed in Fig. 27.

By dividing the muscle activity of the surfing phase by the duration of the phase, relative muscle activity is obtained, as displayed in Fig. 28.

The Take-off phase is measured as the most tiring phase during a wave, even if it is the shortest phase. This is usually mentioned by elite coaches [5–6].

3.2.2. Indicators visualization

It is possible to visualize the four performance indicators along the wave. Board-jerk (BJ), Surfer-jerk (SJ), Muscle Activity (MA) and rotation speed are displayed for wave 4 in Figs. 29–32.

During the top turn (orange phase), the board swept an angle of $\theta_s = 1.66rd$.

3.2.3. Indicators comparison

It is proposed to compare the wave-total proportions represented by the surfing phases in each indicator. The proportions for wave 4 and 5 are presented in Fig. 33.

The first four lines describe wave 4 and the last four describe wave 5. The “Phase duration” lines represent the duration of each phase in the total wave duration. Then each indicator is declined for each wave into the indicator portion for every phase.

For instance, the red portion of “Muscle Activity Wave 4” represents the portion of the MA during the takeoff of wave 4 in the total MA of wave 4.

This visualization aims to detect potential information contained in the data as a quick comparator tool displaying every indicator.

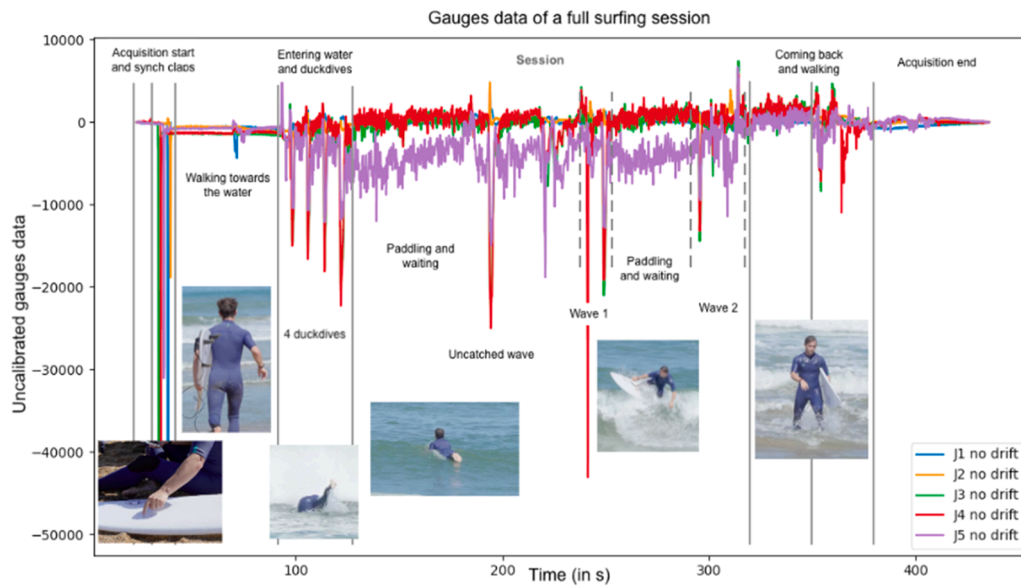


Fig. 17. Gauges data of a full surfing session and the several events that took place during it.

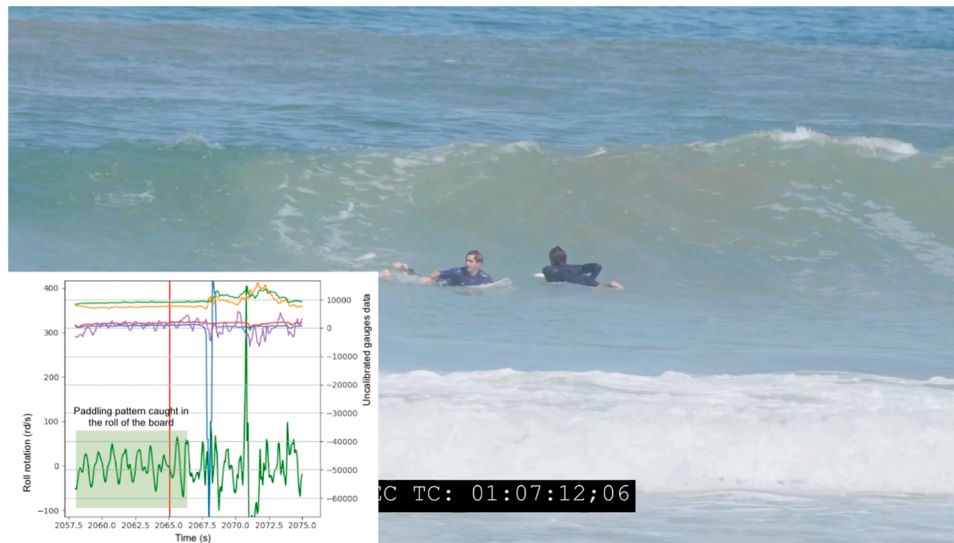


Fig. 18. Paddling pattern caught in the roll measurement of the embedded Arduino's gyroscope.

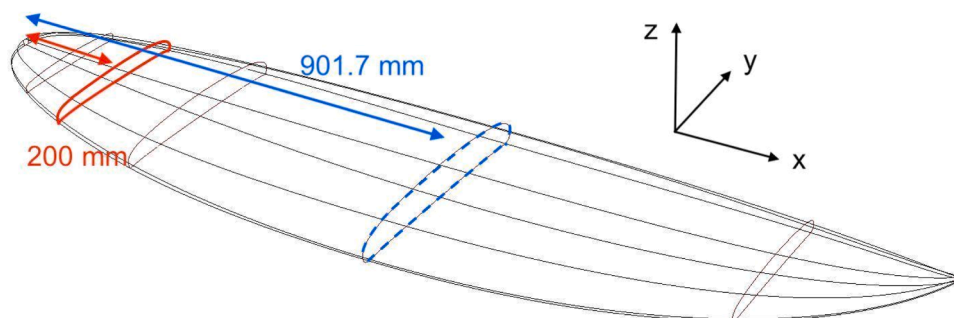


Fig. 19. Position of the displayed slices of the board on Fig. 20. and Fig. 21.

3.2.4. Indicators combination and analysis models

Five models of indicator combinations are presented as potential analysis tools. They are calculated and shown for waves 4 and 5. The first four present discriminators' calculations aiming to describe top-

turn maneuver performance; they consist of different normalizations of the performance indicators to compare them and their interest in the discrimination of top-turns. The last 1 p.m. to describe the quality of surfing on the whole wave.

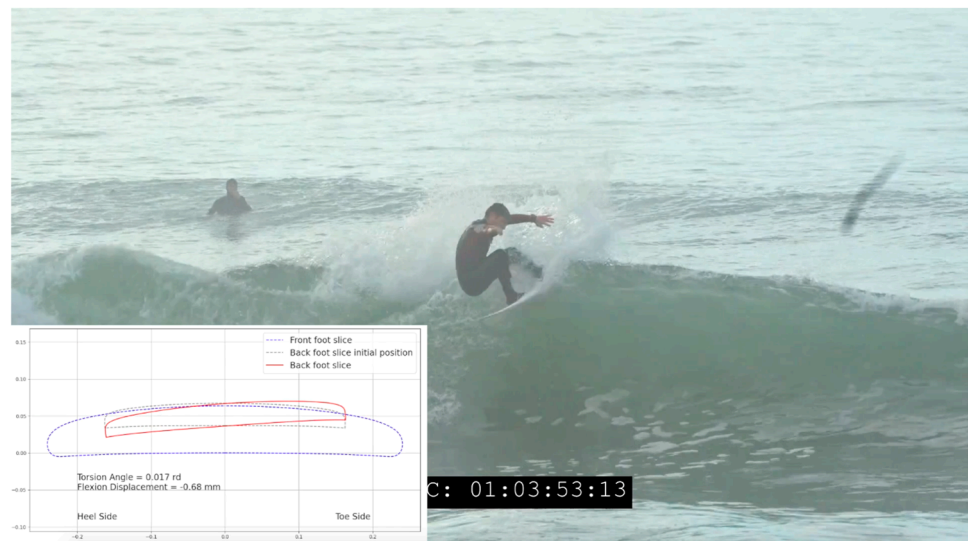


Fig. 20. Torsion and flexion of the board measurement during frontside top-turn re-entry.

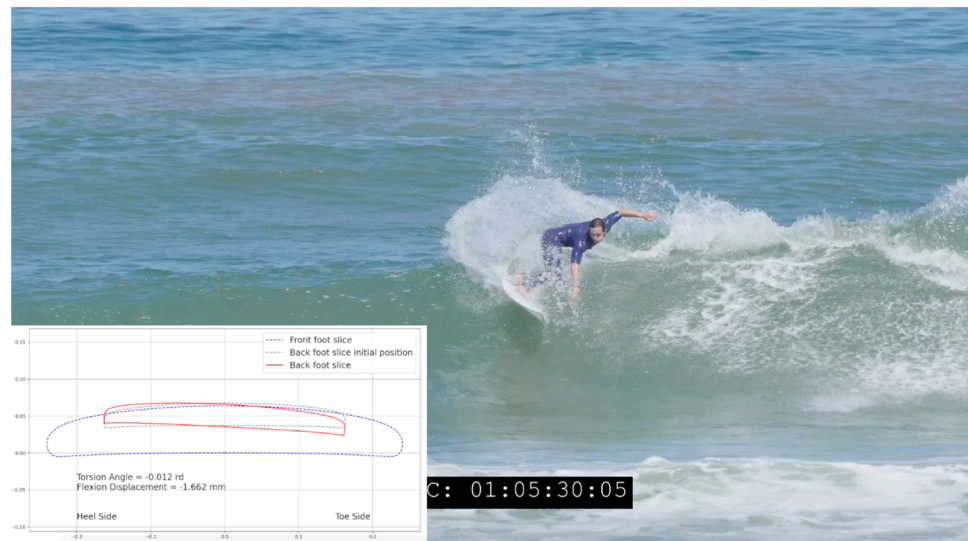


Fig. 21. Torsion and flexion of the board measurement during backside top-turn curve.

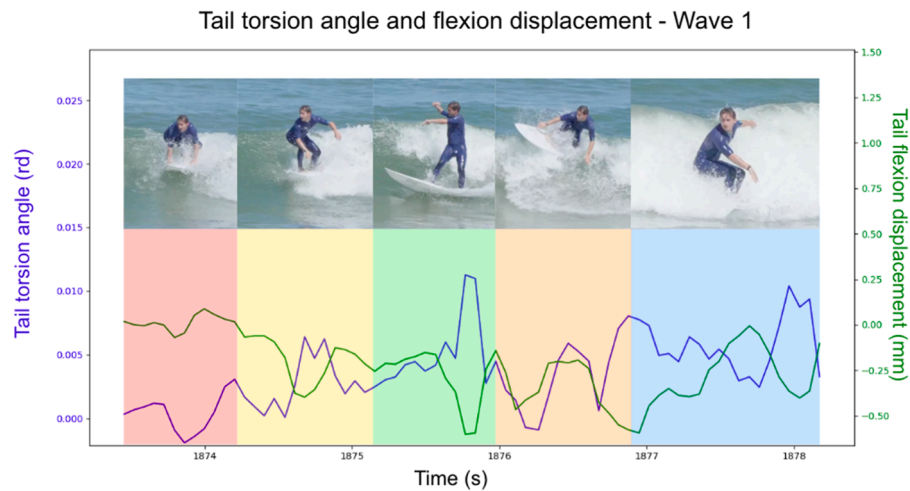


Fig. 22. Torsion and flexion data of the board's tail on wave n°1.

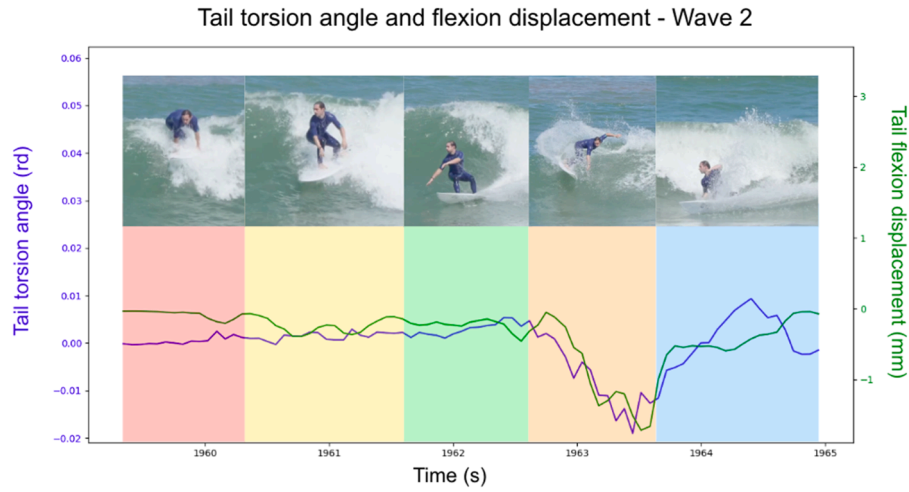


Fig. 23. Torsion and flexion data of the board's tail on wave n°2.

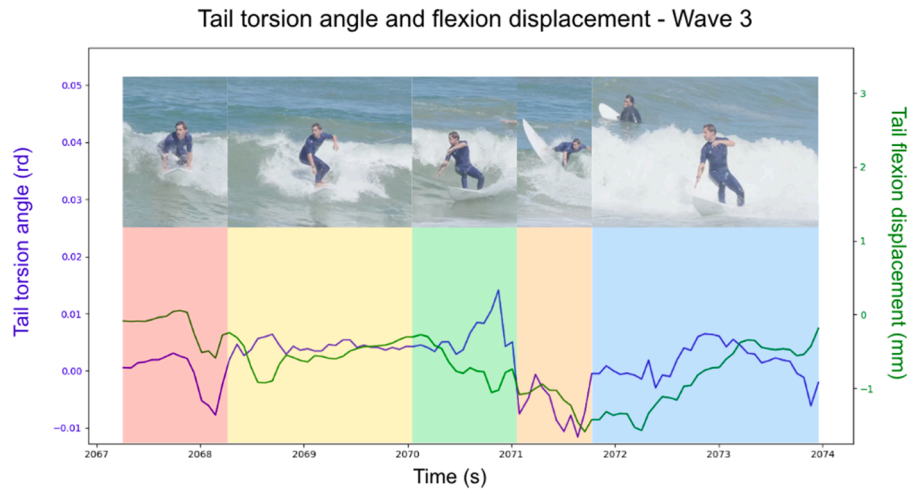


Fig. 24. Torsion and flexion data of the board's tail on wave n°3.

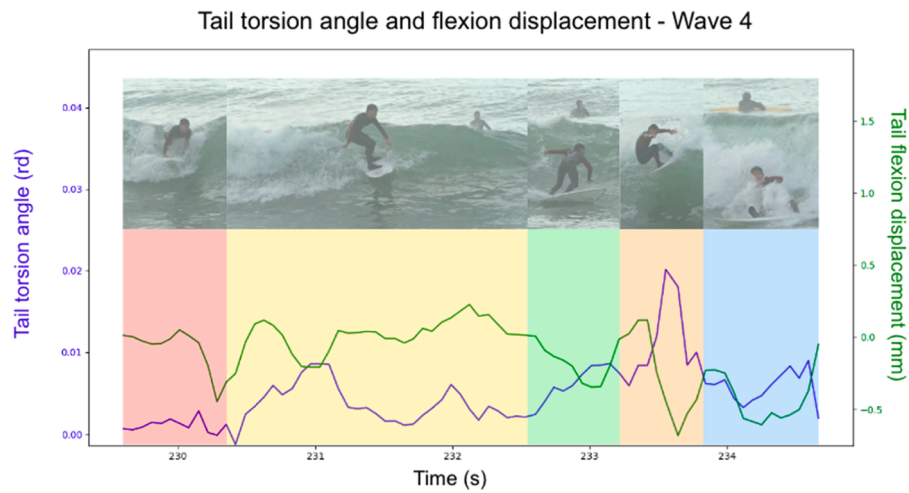


Fig. 25. Torsion and flexion data of the board's tail on wave n°4.

3.2.4.1. Radicality indicator. The Radicality indicator (R_a) of a maneuver is calculated from the Z-axis rotation of the board. To take in consideration both the swept angle and the rotation speed of the board

during the top-turn maneuver, it is proposed to be calculated as so:

$$Ra = \theta_z \cdot \underline{R_z} = \frac{\theta_z^2}{t_1 - t_0} \quad (11)$$

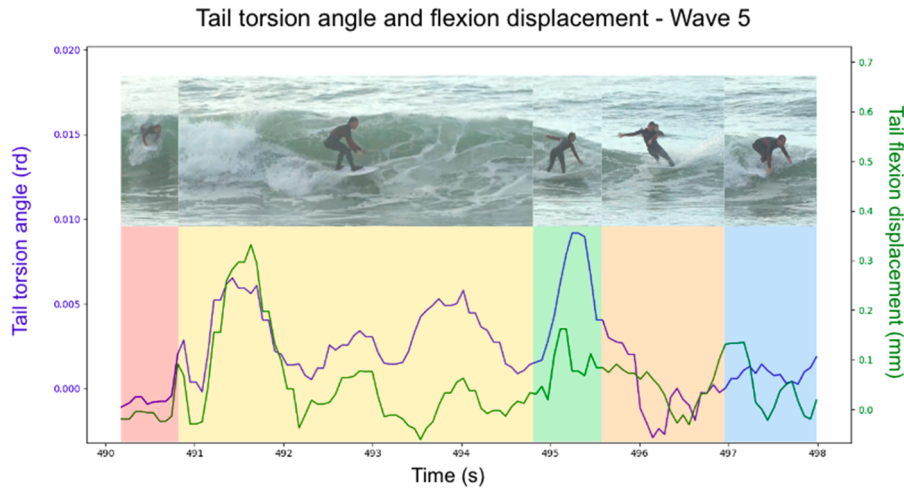


Fig. 26. Torsion and flexion data of the board's tail on wave n°5.

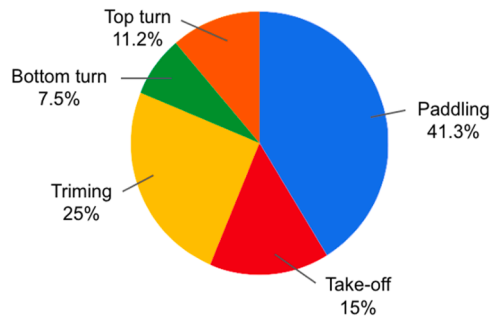


Fig. 27. Muscle activity proportion for each surfing phase.

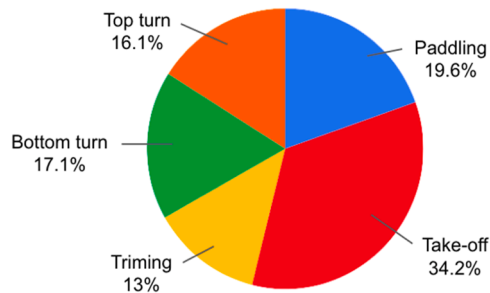


Fig. 28. Relative muscle activity for each surfing phase.

$\bar{R}_z$ is the average rotation speed during the top-turn maneuver, between t_0 and t_1 .

Therefore, the bigger R_a is, the more the maneuver is considered radical.

The R_a calculation of the top-turn maneuver of wave 4 and 5 are presented in Table 4.

3.2.4.2. Raw discrimination. This first discrimination is proposed as a ratio of the raw indicator data and the radicality indicator R_a of the top-turn maneuver. Thus, no comparison must be made between the MA discriminator value and a jerk-based discriminator value.

Let I be an indicator. The raw discrimination ratio is proposed as followed:

$$D_{raw,I} = \frac{I_{Top}}{R_a} \quad (12)$$

The raw discrimination ratios of each indicator for waves 4 and 5 are displayed in Table 5.

3.2.4.3. Bottom turn normalization. The bottom turn and the top turn are two interconnected phases. Every top turn is preceded by a specific bottom turn, putting the surfer and the board in the best possible position for the completion of the top turn maneuver.

It is therefore proposed to normalize top turn indicator quantities by their associated bottom turn quantities.

Let I be an indicator. The normalized indicator I_N is calculated as followed:

$$I_N = \frac{I_{Top}}{I_{Bottom}} \quad (13)$$

The normalized indicator is then compared to the radicality indicator R_a of the maneuver by the discrimination ratio:

$$D_{B,I} = \frac{I_N}{R_a} \quad (14)$$

The normalized indicators and discrimination ratios of each indicator for waves 4 and 5 are displayed in Table 6.

3.2.4.4. Discrimination by muscle activity. This model compares MA with a Jerk value, both bottom-normalized. It aims to compare the effort of the surfer with the amount of sudden moves from the surfer or the board during a maneuver.

Let J be a Jerk indicator. The MA Discriminator is calculated as followed:

$$D_{MA,J} = \frac{MA_N}{J_N} \quad (15)$$

The MA Discriminators for SJ and BJ for wave 4 and 5 are displayed in Table 7.

3.2.4.5. Jerks ratio or piloting ratio. The Piloting Ratio (PR) aims to quantify the correlation between the Surfer-Jerk and the Board-Jerk. It can be applied to full wave jerks and to specific surfing phases or wave sections.

It is defined as followed:

$$PR = \frac{SJ}{BJ} \quad (16)$$

Therefore, if:

- $PR = 1$, the piloting of the surfer would be considered in phase with the reaction of the board, on the surfed wave.

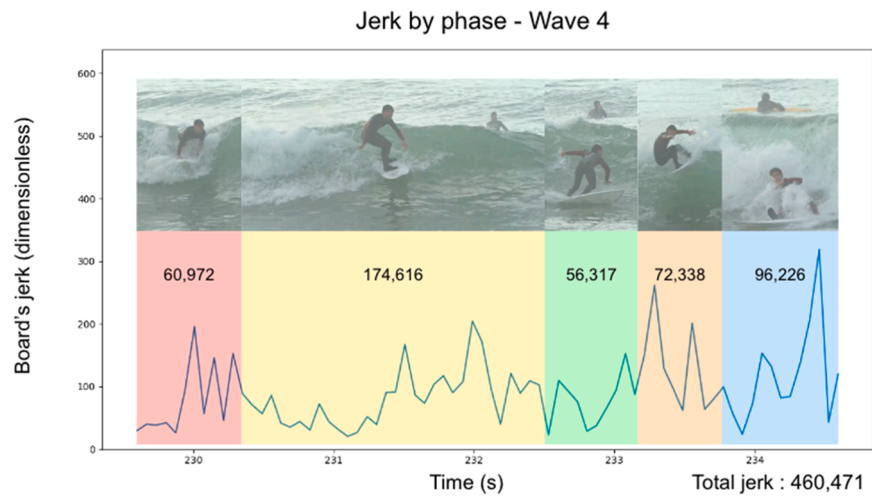


Fig. 29. Board-jerk on wave n°4.

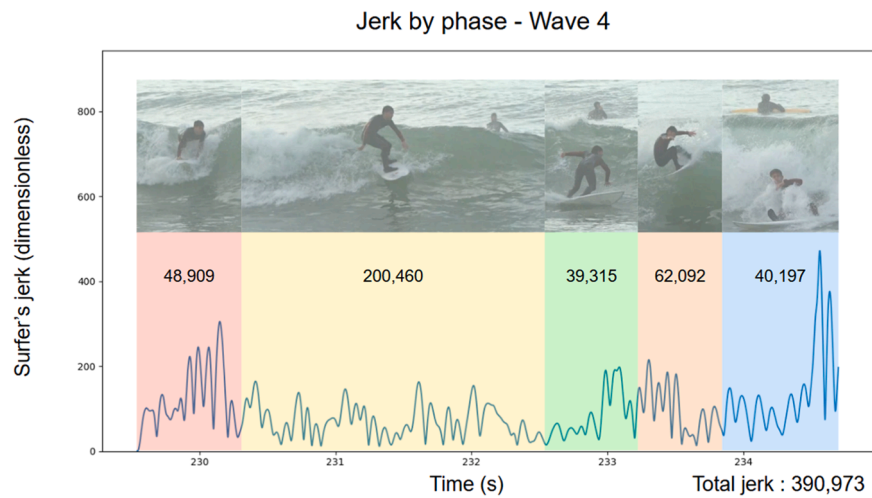


Fig. 30. Surfer-jerk on wave n°4.

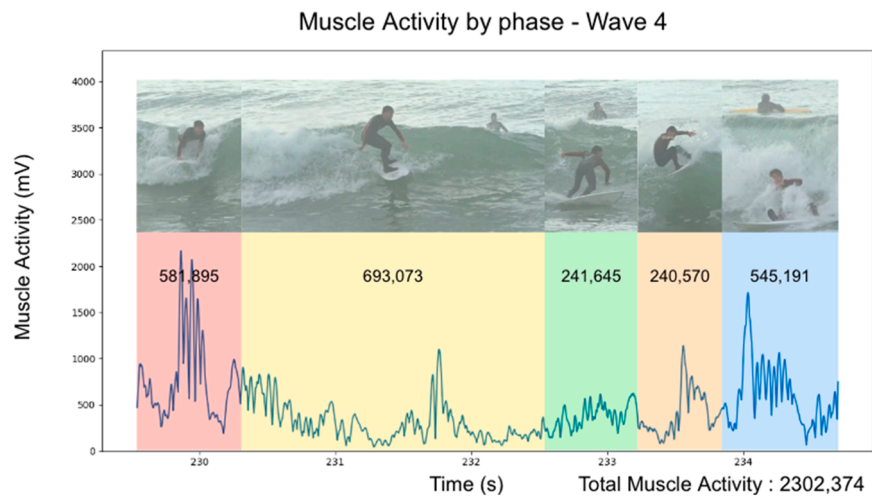


Fig. 31. Muscle Activity on wave n°4.

- $PR > 1$ or $PR < 1$, the piloting of the surfer was made difficult by a bad Surfer-Board-Wave combo.
The PR calculation for waves 4 and 5 is presented in Table 8.

To assess the accuracy of the all-above discriminators and to refine them into their physical meaning, repeated experiments are required. The PR indicator shows interest in its forthcoming physical sense and is a

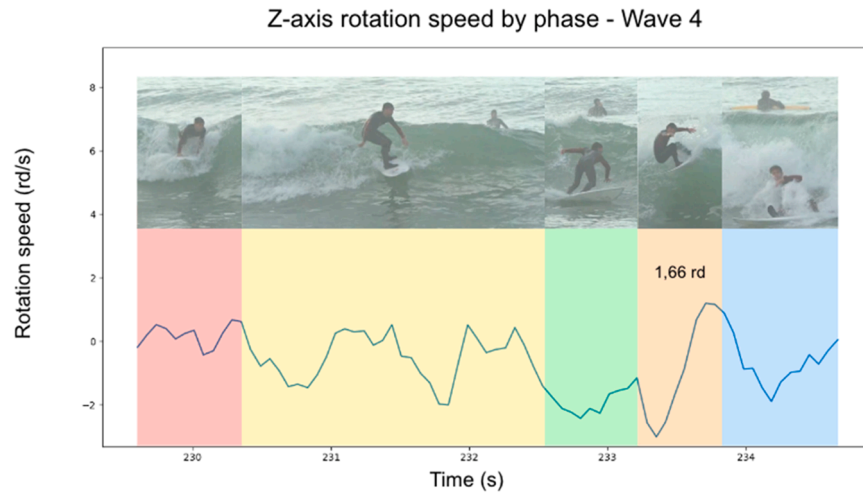
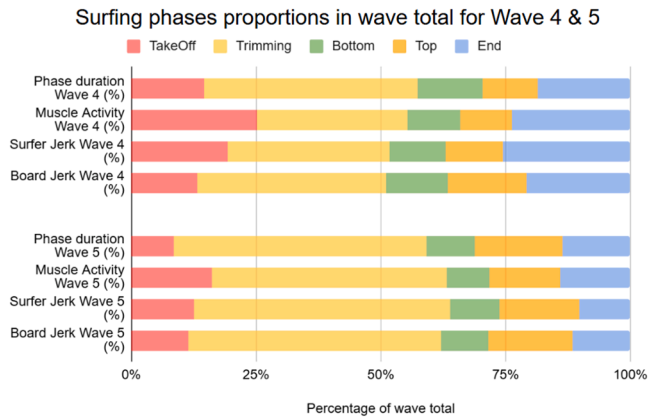
Fig. 32. Rotation speed R_z on wave n° 4.

Fig. 33. Surfing phases proportions in wave total for wave 4 & 5.

Table 4

Radicality Indicator (Ra) calculation for waves 4 & 5 top-turn.

Wave	4	5
Average Rotation Speed (rd/s)	2,94	1,48
Swept Angle (rd)	1,66	2,04
Ra (rd ² /s)	4,89	3,02

Table 5

Raw discriminators calculation for waves 4 & 5 top-turn.

Wave	4	5
Ra	4,89	3,02
$D_{raw,MA}$	49,24	189,95
$D_{raw,SJ}$	11,32	20,59
$D_{raw,BJ}$	14,81	44,35

Table 6

Bottom-normalized Discriminators calculation for waves 4 & 5 top-turn.

Wave	4	5
MA_N	1,00	1,69
SJ_N	1,02	1,58
BJ_N	1,28	1,77
$D_{B,MA}$	0,20	0,56
$D_{B,SJ}$	0,21	0,52
$D_{B,BJ}$	0,26	0,59

Table 7

MA Discriminators calculation for waves 4 & 5 top-turn.

Wave	4	5
$D_{MA,SJ}$	0,98	1,07
$D_{MA,BJ}$	0,79	0,89

Table 8

Piloting Ratio (PR) calculation for whole waves 4 & 5. The PR row turns green as PR gets close to 1 and red the further from 1.

Wave	Phase	Surfer-Jerk	Board-Jerk	PR
4	Takeoff	93,14	60,97	1,53
	Trimming	155,02	174,62	0,89
	Bottom	54,21	56,32	0,96
	Top	55,33	72,34	0,76
	End	122,18	96,23	1,27
	Full Wave	479,87	460,47	1,04
5	Takeoff	48,91	90,13	0,54
	Trimming	200,46	401,50	0,50
	Bottom	39,31	75,41	0,52
	Top	62,09	133,73	0,46
	End	40,20	91,52	0,44
	Full Wave	390,97	792,28	0,49

priority calculation to run on future experiments.

3.2.5. Comparison between indicators and mechanical behavior

In anticipation of future work in a repeated environment, a comparison of the mechanical behavior data and the presented-above indicators and discriminators is proposed.

Let I be an indicator and M be a mechanical behavior (whether flexion displacement noted FD or torsion angle noted TA). It is proposed to consider the ratio:

$$M_I = \frac{M}{I} \quad (17)$$

Those ratios aim to be tools describing in future works the influence of the mechanical characteristics of the board on the overall performance.

The M_I ratios for top-turn maneuver of waves 4 and 5 are presented in Table 9.

A bigger sample size experiment of repeated waves is needed to run statistical tests over comparable data refining the pertinence of these

Table 9M_I ratios calculation for top-turn maneuver of waves 4 & 5.

Wave	M	4		5	
		FD (mm)	TA (rd)	FD (mm)	TA (rd)
I		−0,60 ± 0,15	0,11 ± 0,01	−0,10 ± 0,02	−0,06 ± 0,005
Raw	Ra	−0123	0023	−0033	−0020
	D _{raw} , MA	−0012	0002	−0001	0000
	D _{raw} , SJ	−0053	0010	−0005	−0003
	D _{raw} , BJ	−0041	0007	−0002	−0001
Bottom-Normalized	D _{B,MA}	−2945	0540	−0179	−0107
	D _{B,SJ}	−2872	0527	−0191	−0115
	D _{B,BJ}	−2282	0418	−0170	−0102
MA Discriminator	D _{MA,SJ}	−0615	0113	−0093	−0056
	D _{MA} , BJ	−0755	0138	−0112	−0067
Piloting Ratio	PR _{top}	−0784	0144	−0215	−0129

comparisons. This proposition focuses on merging both mechanical data M and the diverse performance indicators I into ratios to compare. In future work, its interest will be to indicate the most interesting behavior, torsion or flexion, into performance events.

4. Discussion

4.1. Limitations

It is to be noted that this paper presents novel methodical tools to describe surfing performance, and measure mechanical behavior of the tail of a surfboard. The waves presented in this experiment are unique oceanic waves that cannot be duplicated. Thus, no conclusions on surfing performance should be drawn from the in-situ experiment alone presented in this work. The performance indicators and the discriminators must be applied to comparable waves, with the same board and the most similar surfing possible to assert their precision in surfing description and establish consistent connections with the mechanical behavior data.

About measuring mechanical data. The current board design is not sufficient to quantify the mechanical behavior of the board's nose. It was engineered to measure the behavior of the back half of the board supporting the stress of the surfer. If the nose behavior is required for future work with the same method, other strain gauges should be added further on the nose and similar calibration protocols as 2.2.3.2. and 2.2.3.3 should be applied and focused on the nose. It must also be noted that the calibration protocol 2.2.3.1 cannot be applied to real condition data because the boundary condition of the three-point flexion bench does not represent the real condition stress inputs.

About the tail's mechanical measurement. The precision from the current board design (24.6 % in flexion, 8.9 % in torsion) implies possible overlapping of different maneuvers data and can cause analysis biases. Trends should only be concluded after a statistical analysis of larger sample size of repeated waves to compensate for the uncertainty of the measurement.

4.2. Future work

The presented method should be applied to a future experiment. This experiment would take place in a controlled environment with state-of-the-art reproducible waves.

An experienced surfer with an 8/8 technical level would be asked to execute similar turns on two boards of the same shape but different construction.

A significant number of waves will be recorded and measured for each board construction.

The performance quantification method and the tools developed in this work would be applied to identify major performance events, refined by statistical analysis.

The muscle activity indicator will be updated to the energy consumption of the surfer as a better indicator of the smoothness of the combo surfer-board-wave.

In a repeated experiment, it would also be possible to measure and compare similar physiological data of surfing such as full body movements.

The radicality indicator based on z-rotation speed is the first proposal of an indicator to describe a top-turn maneuver. The radicality indicator can be updated with sensor orientation to get a three directional value of the angle swept by the board during the maneuver.

A new method of measuring the board's mechanical characteristics would be tested to get the behavior of the full board.

5. Conclusion

Surfing is a complex physical phenomenon. To get a better understanding of this phenomenon, an essential decomposition is presented: identifying what is fixed, what can be modified, and the outcome to measure. In this work, the focus is made on the development of a multi-sensor surfboard design and a novel performance quantification framework proof-of-concept.

Two challenges were addressed: first, it was required to know the mechanical behavior of the board while surfing. To do so, a board instrumented with strain gauges was designed and manufactured. Calibration benches, analytic computing tools, and data processing were made and written. The current measurement technology and calibration give us the flexion and torsion behavior of the tail in real condition. A second method is to be tested in future work to get the behavior of the full board. In terms, it is expected to equip several flex-different boards with this technology and measure their behavior during surfing.

The second challenge, a way more general one, is to measure the performance of the combo surfer-board-wave. Nowadays, no quantifiable indicator exists to define surfing performance. To apply scientific tools to surfing, performance definition and performance quantification must be the next priority of surfing science.

This work has proposed different performance indicators to decrypt the complex definition of surfing performance: the jerk is introduced, both as board-jerk and surfer-jerk. The muscle activity of the surfer is also measured as an indicator. It is proposed in this work to use the rotation speed of the board and the angle range of the maneuvers as a radicality indicator.

To get the most information out of them, different comparison calculations are proposed, whether by ratios or normalization. In future work, these discriminators should be refined by a larger data sample and statistical calculations.

A model experiment has been set up to get data on surfing phenomena. 5 waves have been recorded in real oceanic waves with the instrumented board including 2 with a fully instrumented surfer. These waves illustrate this novel quantification method along this work.

In further work, the surfer must be a professional surfer, and he/she must surf the different flex boards in a wave pool where the waves are constant and always identical.

A new method of measuring mechanical characteristics is proposed to get the behavior of the entire board and will be tested in-situ in the upcoming works.

CRedit authorship contribution statement

P-E. Charles: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **P. Algourdin:** Methodology. **B. Ostre:** Writing – review & editing,

Validation, Supervision, Project administration, Methodology. **Y. Joliff:** Writing – review & editing, Validation, Supervision, Project administration, Methodology. **L. Bouffier:** Resources. **S. Fleury:** Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.rineng.2025.108868](https://doi.org/10.1016/j.rineng.2025.108868).

Data availability

Data will be made available on request.

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